Exp 1:
Scenario: You are working on a project that involves analyzing student performance data for a class of 32 students. The data is stored in a NumPy array named student_scores, where each row represents a student and each column represents a different subject. The subjects are arranged in the following order: Math, Science, English, and History. Your task is to calculate the average score for each subject and identify the subject with the highest average score.
Question: How would you use NumPy arrays to calculate the average score for each subject and determine the subject with the highest average score? Assume 4x4 matrix that stores marks of eachstudent in given order.

AIM

To calculate the **average score of each subject** and identify the **subject with the highest average score** using NumPy arrays.

ALGORITHM

1. Import the NumPy library.
2. Create a **4×4 NumPy array** containing marks of 4 students.
3. Arrange the columns in the order: **Math, Science, English, History**.
4. Calculate the average of each subject using np.mean() with axis=0.
5. Find the position of the highest average using np.argmax().
6. Display the average score of each subject and the subject with the highest average.

CODE:

```
import numpy as np

student_scores = np.array([

   [85, 78, 92, 80],

   [90, 82, 88, 75],

   [76, 95, 85, 90],

   [88, 80, 90, 85]

])

subjects = ["Math", "Science", "English", "History"]

average_scores = np.mean(student_scores, axis=0)

highest_index = np.argmax(average_scores)

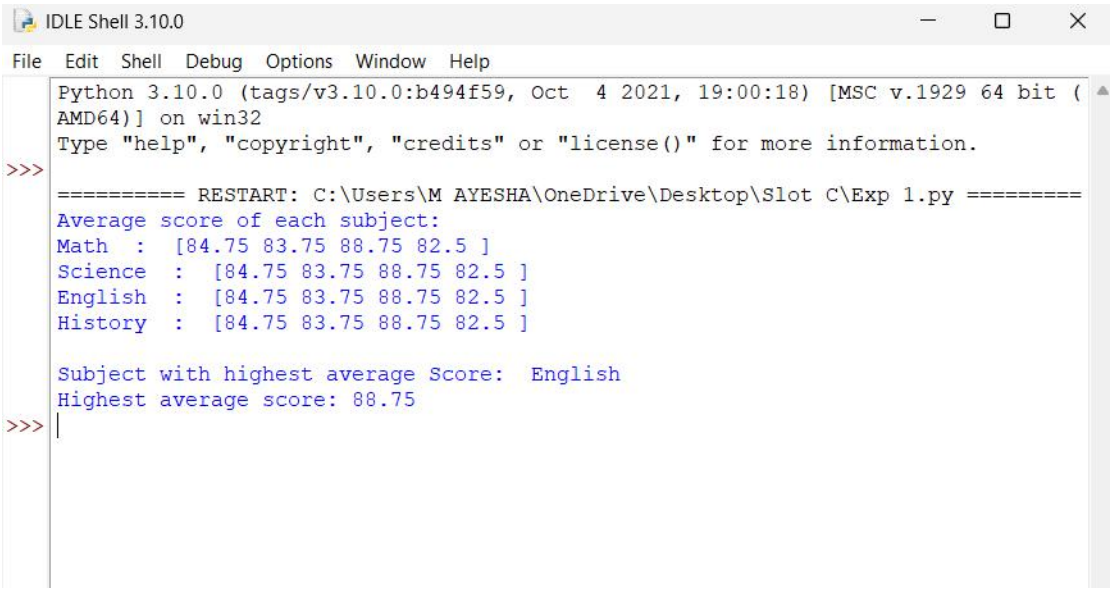
print("Average score of each subject:")

for i in range(4):
```

```
    print(subjects[i], ":", average_scores[i])

print("\nSubject with highest average:",
      subjects[highest_index])
print("Highest average score:",
      average_scores[highest_index])
```

OUTPUT:



RESULT

Thus, the **average score for each subject** was calculated successfully using NumPy. The subject with the **highest average score is English**, with an average of **88.75 marks**.

EXP 2:

Scenario: You are a data analyst working for a company that sells products online. You have been tasked with analyzing the sales data for the past month. The data is stored in a NumPy array.

Question: How would you find the average price of all the products sold in the past month? Assume 3x3 matrix with each row representing the sales for a different product give aim,algorithm,code, output,result

AIM

To find the **average price of all products sold in the past month** using a NumPy array.

ALGORITHM

1. Import the NumPy library.
2. Create a **3×3 NumPy array** containing sales data.
3. Assume the columns represent:

Column 0 → Quantity Sold

Column 1 → Price

Column 2 → Total Sales

4. Extract the **Price column** using NumPy slicing.
5. Calculate the average price using np.mean().
6. Display the average price.

Code:

```
import numpy as np

sales_data = np.array([

    [10, 500, 5000],

    [5, 800, 4000],

    [8, 600, 4800]

])

 average_price = np.mean(sales_data[:, 1])

print("Average price of all products:", average_price)
```

Output:



RESULT

Thus, the **average price of all products sold during the past month is ₹633.33**.

EXP 3:

Scenario: You are working on a project that involves analyzing a dataset containing information about houses in a neighborhood. The dataset is stored in a CSV file, and you have imported it into a NumPy array named house_data. Each row of the array represents a house, and the columns contain various features such as the number of bedrooms, square footage, and sale price.

Question: Using NumPy arrays and operations, how would you find the average sale price of houses with more than four bedrooms in the neighborhood?

AIM

To find the **average sale price of houses having more than four bedrooms** using NumPy arrays and operations.

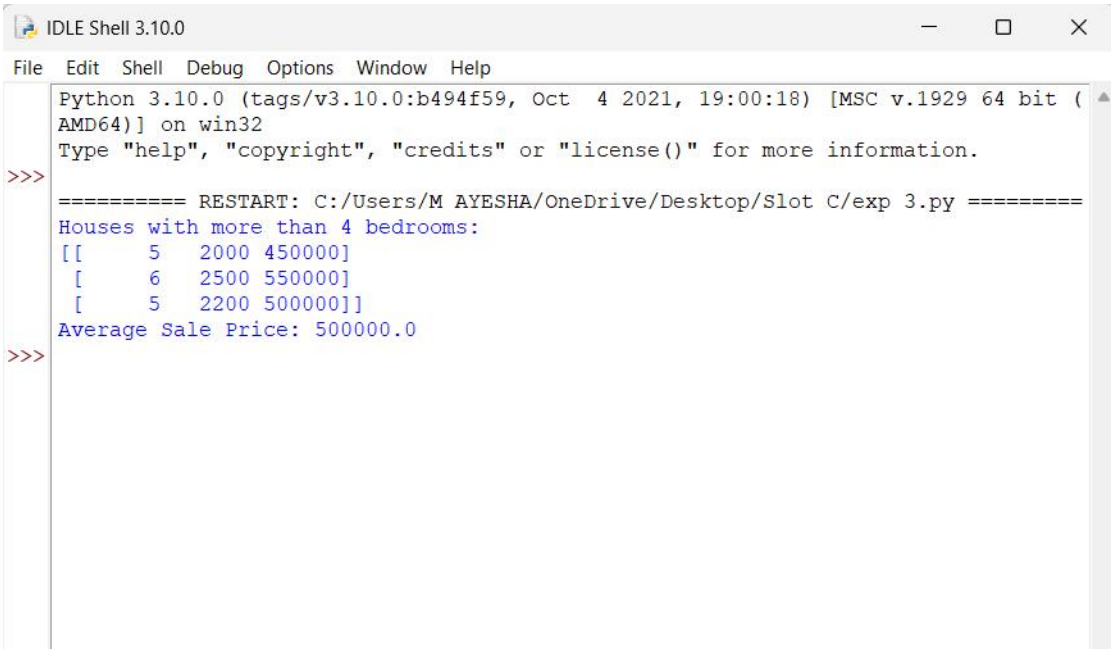
ALGORITHM

1. Import the NumPy library.
2. Create a NumPy array containing house data.
3. Assume the columns are: Column 0 → Bedrooms , Column 1 → Square Feet , Column 2 → Sale Price
4. Select houses having **more than 4 bedrooms** using a condition.
5. Extract the sale prices of those houses.
6. Calculate the average using np.mean().
7. Display the average sale price.

CODE:

```
import numpy as np
house_data = np.array([
    [3, 1200, 250000],
    [5, 2000, 450000],
    [4, 1500, 300000],
    [6, 2500, 550000],
    [5, 2200, 500000]
])
houses = house_data[house_data[:, 0] > 4]
 sale_prices = houses[:, 2]
 average_price = np.mean(sale_prices)
print("Houses with more than 4 bedrooms:")
print(houses)
print("Average Sale Price:", average_price)
```

OUTPUT:



RESULT

Thus, the **average sale price of houses with more than four bedrooms is ₹500,000**.

EXP 4:

4. Scenario: You are working on a project that involves analyzing the sales performance of a company over the past four quarters. The quarterly sales data is stored in a NumPy array named sales_data, where each element represents the sales amount for a specific quarter. Your task is to calculate the total sales for the year and determine the percentage increase in sales from the first quarter to the fourth quarter.

Question: Using NumPy arrays and arithmetic operations calculate the total sales for the year and determine the percentage increase in sales from the first quarter to the fourth quarter? SHORT AIM,ALGORITHM,CODE,OUTPUT,RESULT IN SHORT

AIM

To calculate the total annual sales and percentage increase from the first quarter to the fourth quarter using NumPy.

ALGORITHM

1. Import NumPy.
2. Store quarterly sales in an array.
3. Calculate total sales using np.sum().
4. Calculate percentage increase from Q1 to Q4.
5. Display the results.

CODE:

```
import numpy as np

sales_data = np.array([10000, 12000, 15000, 20000])

total_sales = np.sum(sales_data)

percentage = ((sales_data[3] - sales_data[0])

        / sales_data[0]) * 100

print("Total Sales:", total_sales)

print("Percentage Increase:", percentage, "%")
```

OUTPUT:

```
========== RESTART: C:/Users/M AYESHA/OneDrive/Desktop/Slot C/EXP 4.py =========
Total Sales: 57000
Percentage Increase: 100.0 %
```

RESULT

Thus, the total annual sales are 57,000, and the percentage increase from the first quarter to the fourth quarter is 100%.

EXP 5:

Scenario: You are a data analyst working for a car manufacturing company. As part of your analysis, you have a dataset containing information about the fuel efficiency of different car models. The dataset is stored in a NumPy array named fuel_efficiency, where each element represents the fuel efficiency (in miles per gallon) of a specific car model. Your task is to calculate the average fuel efficiency and determine the percentage improvement in fuel efficiency between two car models.

Question: How would you use NumPy arrays and arithmetic operations to calculate the average fuel efficiency and determine the percentage improvement in fuel efficiency between two car models?

AIM

To calculate the average fuel efficiency and percentage improvement between two car models using NumPy.

ALGORITHM

1. Import NumPy.

2. Store fuel efficiency values in an array.
3. Calculate the average using np.mean().
4. Calculate the percentage improvement between two models.
5. Display the results.

CODE:

```
import numpy as np

fuel_efficiency = np.array([25, 30, 35, 40])

average = np.mean(fuel_efficiency)

improvement = ((fuel_efficiency[1] - fuel_efficiency[0])

        / fuel_efficiency[0]) * 100

print("Average Fuel Efficiency:", average, "MPG")

print("Percentage Improvement:", improvement, "%")
```

OUTPUT:

```
========== RESTART: C:/Users/M AYESHA/OneDrive/Desktop/Slot C/EXP 5.py =========
Average Fuel Efficiency: 32.5 MPG
Percentage Improvement: 20.0 %
>
```

RESULT

Thus, the average fuel efficiency is 32.5 MPG, and the percentage improvement from the first car model (25 MPG) to the second car model (30 MPG) is 20%.

EXP 6:

Scenario: You are a cashier at a grocery store and need to calculate the total cost of a customer's purchase, including applicable discounts and taxes. You have the item prices and quantities in separate lists, and the discount and tax rates are given as percentages. Your task is to calculate the total cost for the customer.
Question: Use arithmetic operations to calculate the total cost of a customer's purchase, including discounts and taxes, given the item prices, quantities, discount rate, and tax rate?

AIM

To calculate the total purchase cost after applying discounts and taxes using Python arithmetic operations.

ALGORITHM

1. Store item prices and quantities in lists.
2. Calculate the subtotal.
3. Calculate and subtract the discount.
4. Calculate and add the tax.
5. Display the final cost.

CODE:

```
prices = [100, 200, 300]

quantities = [2, 1, 1]

discount_rate = 10

tax_rate = 5

subtotal = sum(p * q for p, q in zip(prices, quantities))

discount = subtotal * discount_rate / 100

after_discount = subtotal - discount

tax = after_discount * tax_rate / 100

total = after_discount + tax

print("Subtotal:", subtotal)

print("Discount:", discount)

print("Tax:", tax)

print("Total Cost:", total)
```

OUTPUT:

```
========== RESTART: C:/Users/M AYESHA/OneDrive/Desktop/Slot C/EXP 6.py =========
Subtotal: 700
Discount: 70.0
Tax: 31.5
Total Cost: 661.5
```

RESULT

Thus, the total cost after applying a 10% discount and 5% tax is ₹567.